

作业1:

$$1 \quad (a) \quad \bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$$

$$Error(\bar{x}) = \frac{1}{N} \sum_{i=1}^N \mathcal{E}(x_i)$$

$$Max(Error(\bar{x})) = \frac{1}{N} Max\left(\sum_{i=1}^N \mathcal{E}(x_i)\right)$$

$$= \frac{1}{N} \sum_{i=1}^N Max(\mathcal{E}(x_i))$$

$$= \frac{1}{N} \sum_{i=1}^{N-1} \frac{1}{2} \mathcal{E}_M(i+1)$$

$$= \frac{1}{N} \cdot \frac{1}{2} \mathcal{E}_M \frac{(N+2)(N-1)}{2} = \frac{(N+2)(N-1)}{2N} \frac{\mathcal{E}_M}{2} \max |x_i|$$

$$\sim \frac{N}{2} \frac{\mathcal{E}_M}{2} \max |x_i|$$

(b) 等价性证明:

$$\sum_{i=1}^N (x_i - \bar{x})^2 = \sum_{i=1}^N (x_i^2 + \bar{x}^2 - 2x_i\bar{x}) = \sum_{i=1}^N x_i^2 + N \cdot \bar{x}^2 - 2 \sum_{i=1}^N x_i \bar{x}$$

$$= \sum_{i=1}^N x_i^2 + N \cdot \bar{x}^2 - 2N \cdot \bar{x}^2 = \sum_{i=1}^N x_i^2 - N \cdot \bar{x}^2$$

$$\therefore S^2 = \frac{1}{N-1} \left( \sum_{i=1}^N x_i^2 - N \bar{x}^2 \right) = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2$$

$$\text{误差: } \mathcal{E}(S^2) = \frac{1}{N-1} \mathcal{E}\left(\sum_{i=1}^N x_i^2 - N \bar{x}^2\right)$$

$$\sim \frac{1}{N} \mathcal{E}\left(\sum_{i=1}^N x_i^2\right) + \mathcal{E}(\bar{x}^2)$$

$$\sim \frac{1}{N} \sum_{i=1}^N 2x_i \mathcal{E}(x_i) + 2\bar{x} \mathcal{E}(\bar{x})$$

$$\sim 2 \cdot \bar{x} \cdot \frac{\mathcal{E}_M}{2} + 2\bar{x} \cdot \frac{1}{N} \mathcal{E}_M$$

$$\sim \bar{x} \cdot \mathcal{E}_M$$

$$E(S_e^2) = E\left(\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2\right)$$

$$= \frac{1}{N-1} \sum_{i=1}^N 2(x_i - \bar{x}) [E(x_i) + E(\bar{x})]$$

$$= \frac{1}{N-1} \sum_{i=1}^N 2(x_i - \bar{x}) \frac{E_N}{2} \left(1 + \frac{1}{\sqrt{N}}\right)$$

$$\sim \frac{1}{N} \cdot 2 \cdot \sqrt{\text{var}(X)} \cdot \frac{E_N}{2}$$

$$\sim \frac{1}{N} \sqrt{\text{var}(X)} E_N$$

$$\frac{E(S_e^2)}{E(S_e^2)} \propto \frac{\bar{x} E_N}{\frac{1}{N} \sqrt{\text{var}(X)} E_N} = \frac{N \bar{x}}{\sqrt{\text{var}(X)}}$$

因此, 若  $\{x_i\}$  数据的数值较大 (  $\bar{x}$  较大 ) 且  $\{x_i\}$  数值较为集中 (  $\text{var}(X)$  较小 )

则 第一种方法的误差较大, 应使用第二种方法

反之, 若  $\{x_i\}$  数据的数值较小, 集中于中, 且  $\{x_i\}$  数值方差较大, 则应选择

第二种方法.

(c) 积分: 
$$I_n = \int_0^1 dx \left( \frac{x^n}{x+5} \right)$$

验证递推关系:

$$n=0, \quad I_n = \int_0^1 dx \frac{1}{x+5} = \ln(x+5) \Big|_0^1 = \ln\left(\frac{6}{5}\right)$$

$$I_k + 5I_{k-1} = \int_0^1 dx \frac{x^k}{x+5} + \int_0^1 dx \frac{5x^{k-1}}{x+5}$$

$$= \int_0^1 dx \frac{\cancel{(x+5)} x^{k-1}}{\cancel{x+5}}$$

$$= \int_0^1 dx x^{k-1} = \frac{1}{k}$$

递推关系得证.

若  $I_0$  有微小误差:

$$121 \quad I_k = \frac{1}{k} - 5I_{k-1}$$

$$= \frac{1}{k} - 5\left(\frac{1}{k-1} - 5I_{k-2}\right)$$

$$= \sum_{i=1}^k \frac{(-5)^{k-i}}{i} + (-5)^k I_0$$

$$|8I_k| = 5^k |8I_0|$$

当  $k \gg 1$  时.

$|8I_k|$  发散 ( $k \rightarrow \infty$ )  $\therefore I_n$  绝对误差发散.

$$k \rightarrow \infty: \quad \frac{|8I_k|}{|I_k|} = \left| \frac{5^k |8I_0|}{\sum_{i=1}^k \frac{(-1)^{k-i}}{i} + (-5)^k I_0} \right| \rightarrow \frac{5^k}{5^k} \left| \frac{8I_0}{I_0} \right| \quad I_n \text{ 相对误差稳定.}$$

2. (a)  $\det(A) = 1^n = 1 \neq 0$

$\therefore A$  不是奇异矩阵.

(b)  $A = \dots P_{(2,1)} P_{(3,1)} P_{(2,1)} I.$

$$\therefore A^{-1} = P_{(2,1)}^{-1} P_{(3,1)}^{-1} P_{(3,2)}^{-1} \dots = \begin{bmatrix} 1 & 2 & 2 & 2 \\ & 1 & 2 & 2 \\ & & 1 & 2 \\ 0 & & & 1 \end{bmatrix}$$

$$(A^{-1})_{ij} = \begin{cases} 0 & i > j \\ 1 & i = j \\ 2^{j-i-1} & i < j \end{cases}$$

(c)  $\|A\|_p = \sup_{x \neq 0} \frac{\|Ax\|_p}{\|x\|_p} = \sup_{\|x\|_p=1} \|Ax\|_p.$

给定上界.  $\forall x \quad \|x\|_p = 1 \quad \bar{A}:$

$$\|Ax\|_p \leq n^{1/p} \|Ax\|_\infty$$

$$\therefore \forall x, \quad \|Ax\|_\infty = \max_j \left| \sum_i a_{ij} x_i \right| \leq \max_i \sum_j |a_{ij}| |x_j| \leq \max_i \sum_j |a_{ij}| \cdot \|x\|_\infty$$

$$\text{又 } \|x\|_\infty \leq \|x\|_p = 1 \quad \therefore \|A\|_p = \sup_{\|x\|_p=1} \|Ax\|_p \leq n^{1/p} \max_i \sum_j |a_{ij}|$$

对  $p \rightarrow \infty$  有  $n^{1/p} \rightarrow 1$  因此,

$$\lim_{p \rightarrow \infty} \sup \|A\|_p \leq \max_i \sum_j |a_{ij}|$$

再给下界: 令

$$r := \operatorname{argmax}_i \sum_j |a_{ij}|$$

取向量  $x$  使得  $x_j = \operatorname{sgn}(a_{rj})$  (对所有  $j$ ) 则  $\|x\|_p = n^{1/p}$

$$\text{对该 } x: \|Ax\|_p \geq \|Ax\|_\infty \geq \left| \sum_j a_{rj} x_j \right| = \sum_j |a_{rj}| = \max_i \sum_j |a_{ij}|$$

$$\therefore \text{对于 } \bar{x} = \frac{x}{\|x\|_p} \Rightarrow (\|\bar{x}\|_p = 1)$$

$$\text{有 } \|A\|_p \geq \|A\bar{x}\|_p = \frac{\|Ax\|_p}{\|x\|_p} \geq \frac{\max_i \sum_j |a_{ij}|}{n^{1/p}}$$

$$\text{令 } p \rightarrow \infty. \text{ 右端} \rightarrow \max_i \sum_j |a_{ij}|$$

$$\therefore \text{综上, } \lim_{p \rightarrow \infty} \|A\|_p = \max_{i=1, \dots, n} \sum_{j=1}^n |a_{ij}|$$

$$(d) \text{ 由 } \|A\|_2 = \sqrt{\lambda_{\max}(A^*A)} \Rightarrow \text{对 } \forall x.$$

$$\|Ux\|_2^2 = (Ux)^*(Ux) = x^* U^* U x = x^* x = \|x\|_2^2$$

$$\therefore U^* U = I \quad \therefore \text{对 } \forall x \neq 0. \quad \frac{\|Ux\|_2}{\|x\|_2} = 1$$

$$\text{从定义得: } \|U\|_2 = 1 \quad \text{同理 } \|U^*\|_2 = 1$$

对  $\forall A, \forall x.$

$$\|UAx\|_2 = \|Ax\|_2$$

条件数:

$$\kappa_2(UA) = \|UA\|_2 \|(UA)^T\|_2 = \|A\|_2 \|A^T U^T\|_2$$

$$U^T = U^T \text{ 且 } \|U^T\|_2 = 1$$

$$\text{再平方得 } \|XY\|_2 \leq \|X\|_2 \cdot \|Y\|_2.$$

$$\Rightarrow \|A^T U^T\|_2 = \|A^T\|_2 \|U^T\|_2 = \|A^T\|_2.$$

$$\therefore \kappa_2(UA) = \|A\|_2 \|A^T\|_2 = \kappa_2(A)$$

$$\text{3. (a)} \quad B = A^T A = \begin{pmatrix} 2 & 1 & 2 \\ 1 & 2 & 2 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 2 \\ 2 & 2 \end{pmatrix} = \begin{pmatrix} 9 & 8 \\ 8 & 9 \end{pmatrix}$$

$$|\lambda I - B| = \begin{vmatrix} \lambda - 9 & -8 \\ -8 & \lambda - 9 \end{vmatrix} = (\lambda - 9)(\lambda - 9) - 64 = 0.$$

$$(\lambda - 9)^2 = 64 \quad \lambda = 9 \pm 8 \quad \lambda_1 = 1 \quad \lambda_2 = 17$$

$$\text{特征向量: } \begin{pmatrix} \lambda - 9 & -8 \\ -8 & \lambda - 9 \end{pmatrix} \begin{pmatrix} v_{i1} \\ v_{i2} \end{pmatrix} = 0$$

$$\lambda_1 = 1 \quad \vec{v}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\lambda_2 = 17 \quad \vec{v}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \quad \lambda = \begin{pmatrix} 1 & 0 \\ 0 & 17 \end{pmatrix}$$

$$\text{令 } B = A^T A \quad |B|$$

$$B^T = (A^T A)^T = A^T A = B \quad \text{即 } B^T = B \quad B \text{ 为实对称矩阵}$$

B 可以对角化且特征值非负.

$$b_1 \quad A \vec{v}_1 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$A \vec{v}_2 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 3 \\ 3 \end{pmatrix}$$

$$\vec{u}_1 = \frac{A \vec{v}_1}{\sqrt{\lambda_1}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\vec{u}_2 = \frac{A \vec{v}_2}{\sqrt{\lambda_2}} = \frac{1}{\sqrt{3}} \begin{pmatrix} 3 \\ 3 \end{pmatrix}$$

$$(\vec{u}_1, \vec{u}_2) = 0 \quad \vec{u}_1, \vec{u}_2 \text{ 正交}$$

$$\vec{u}_3 = \vec{u}_1 \times \vec{u}_2 = \frac{1}{2\sqrt{17}} \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 0 \\ 3 & 3 & 4 \end{vmatrix} = \frac{1}{2\sqrt{17}} \begin{pmatrix} -4 \\ 4 \\ 3 \end{pmatrix} = \frac{1}{\sqrt{17}} \begin{pmatrix} -2 \\ 2 \\ 3 \end{pmatrix}$$

$$\therefore \vec{u}_1 = \frac{1}{\sqrt{2}} (1, -1, 0)^T$$

$$\vec{u}_2 = \frac{1}{\sqrt{34}} (3, 3, 4)^T$$

$$\vec{u}_3 = \frac{1}{\sqrt{17}} (-2, 2, 3)^T$$

$$4. \text{ 证毕. } A = \sqrt{\lambda_1} \vec{u}_1 \vec{v}_1^T$$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}$$

$$\|A - A_1\|_F = \left\| \begin{pmatrix} \frac{3}{2} & \frac{3}{2} \\ \frac{3}{2} & \frac{3}{2} \end{pmatrix} \right\|_F = \sqrt{\frac{9}{4} \times 4 + 4 \times 2} = 1 = \lambda_2$$

$$\text{发现 } \|A - A_1\|_F = \lambda_2. \quad \text{验证成功}$$

$$4. \quad a) \quad \forall x \neq 0.$$

$$x^T A x = \sum_{i=1}^N \sum_{j=1}^N a_{ij} x_i x_j = \sum_{i=1}^N a_{ii} x_i^2 + \sum_{i \neq j} \sum_{j=1}^N a_{ij} x_i x_j$$

$$a_{ij} x_i x_j \geq -|a_{ij}| |x_i| |x_j|$$

$$\text{由 } |x_i| |x_j| \leq \frac{1}{2} (x_i^2 + x_j^2)$$

$$\Rightarrow x^T A x \geq \sum_{i=1}^N a_{ii} x_i^2 - \sum_{j=1}^N \sum_{i \neq j} \frac{1}{2} |a_{ij}| x_j^2$$

$$x^T A x \geq \sum_{j=1}^N \left[ a_{jj} - \sum_{i \neq j} \frac{1}{2} (|a_{ji}| + |a_{ij}|) \right] x_j^2$$

$$\text{又: } a_{jj} > 0, \quad |a_{ji}| > \sum_{i \neq j} |a_{ij}|$$

$$\therefore a_{jj} - \sum_{i \neq j} \frac{1}{2} (|a_{ji}| + |a_{ij}|) > 0.$$

$$\therefore x^T A x > 0 \quad A \text{ 为正定阵}$$

$$b) \quad K \text{ 正定} \Leftrightarrow K \text{ 的所有顺序主子式} > 0.$$

$$d_1 = 3 > 0$$

$$d_2 = \det \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix} = 8 > 0.$$

$$d_3 = \det \begin{pmatrix} 3 & -1 & c \\ -1 & 3 & 0 \\ c & 0 & 3 \end{pmatrix} = 27 - 3c^2 - 3 = 3(8 - c^2)$$

$$d_4 = \det(M) = d_3 = 3(8 - c^2)$$

$$\text{当 } c = 1 \text{ 时} \quad 3(8 - c^2) = 21 > 0. \quad \therefore K \text{ 正定.}$$

$$\text{若要求 } K \text{ 正定, 则 } 3(8 - c^2) > 0.$$

$$8 > c^2 \quad -2\sqrt{2} < c < 2\sqrt{2}$$

$$\therefore c \text{ 的取值范围为 } (-2\sqrt{2}, 2\sqrt{2})$$

$$5. \quad f(x) = \exp(5x)$$

$$f(x) \approx \frac{f(x+h) - f(x)}{h} \approx f'(x) + \frac{f''(x)}{2} h + O(h^2)$$

$$\text{截断误差为: } E_{\text{trunc}} = \frac{f''(x)}{2} h$$

$$f'(x) = 25e^{5x} \quad f''(x=-1) = 25e^{-5}$$

舍入误差:  $E_{\text{round}} = \frac{E_{\text{in}} f(x)}{h}$

对  $\alpha = -1$ :  $E_{\text{round}} = \frac{E_{\text{mach}} e^{-5}}{h}$

∴ 总误差:

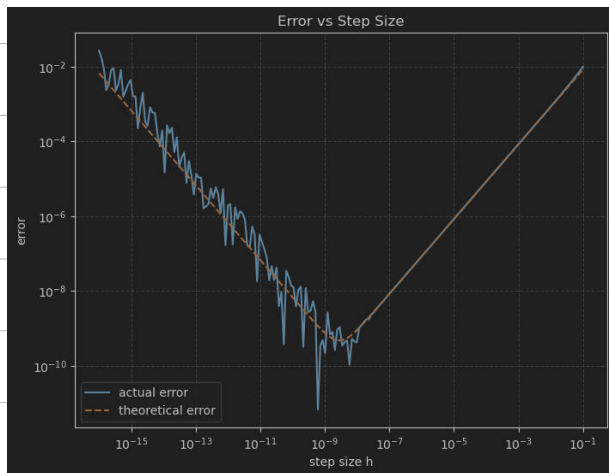
$$E_{\text{tot}} \approx \frac{25}{2} e^{-5} h + \frac{E_{\text{mach}} e^5}{h}$$

最优步长选取:

$$h_{\text{optimal}} = \sqrt{\frac{2 E_{\text{mach}}}{25}}$$

在双精度  $E_{\text{mach}} \sim 10^{-16}$  下:

$$h_{\text{optimal}} \sim \sqrt{\frac{2 \times 10^{-16}}{25}} \approx 3 \times 10^{-9}.$$





6

通项:

$$a_k = \frac{1}{k(k+x)}$$

$$\text{当 } k \text{ 很大} \quad a_k \sim \frac{1}{k^2} \quad \text{且} \quad a_k < \frac{1}{k^2}$$

⇒ 单调递减正项级数, 利用积分法近似尾项误差.

$$R_N = \sum_{k=N+1}^{\infty} a_k \leq \int_N^{\infty} \frac{dx}{k(k+x)}$$

$$\Rightarrow R_N = \frac{1}{x} \ln \frac{N+x}{N}$$

$$\text{要求 } R_N < 10^{-6} \quad \text{于是}$$

$$\frac{1}{x} \ln \frac{N+x}{N} < 10^{-6}$$

$$\text{当 } x=0 \text{ 时: } a_k = \frac{1}{k^2} \quad \text{尾项误差 } R_N = \int_N^{\infty} \frac{dx}{k^2} = \frac{1}{N} \quad \text{此时 } N > 10^6$$

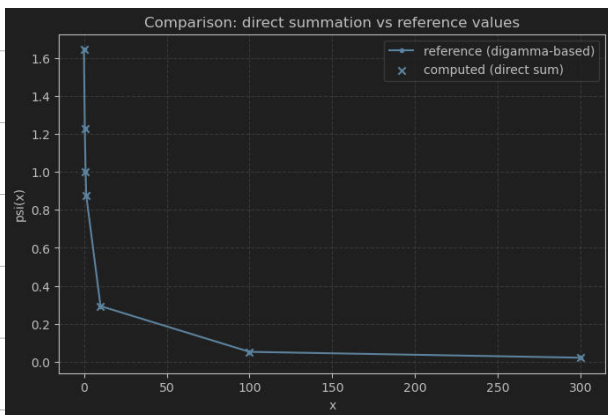
取  $N = 10^6$  即可.

算法:

1. 设置容差  $\varepsilon$

2. 累加至尾项  $R_N < \varepsilon$ .

3. 对每个  $x$  输出对应的  $\eta(x)$



代码与结果

```
1 import math
2
3 def psi_basic(x, eps=1e-6):
4     # 对 x = 0 特殊处理
5     if x == 0:
6         N = int(1/eps) + 1
7         s = 0.0
8         for k in range(1, N+1):
9             s += 1 / (k*k)
10        return s
11
12    # 非零 x 情况
13    s = 0.0
14    k = 1
15    while True:
16        s += 1 / (k * (k + x))
17        # 估计余项误差
18        Rn = (1/x) * math.log((k + x) / k)
19        if Rn < eps:
20            break
21        k += 1
22    return s
23
24 # 测试输入
25 xs = [0.0, 0.5, 1.0, math.sqrt(2), 10.0, 100.0, 300.0]
26
27 for i, x in enumerate(xs, start=1):
28     val = psi_basic(x)
29     print(f"{i:2d} x = {x}, psi(x) = {val:.15f}")
30
```

Executed at 2025.10.04 22:41:27 in 1s 498ms

```
✓ 1 x = 0.0, psi(x) = 1.644933066849770
2 x = 0.5, psi(x) = 1.227410277760964
3 x = 1.0, psi(x) = 0.999999000001048
4 x = 1.4142135623730951, psi(x) = 0.874981996022131
5 x = 10.0, psi(x) = 0.292895825398311
6 x = 100.0, psi(x) = 0.051872775177896
7 x = 300.0, psi(x) = 0.020941212935823
```